

	difference remain the same as object falls, upthrust will remain unchanged.
8	C When 3.0 N load is placed, $F = kx$, $3.0 = k(0.030)$, so $k = 100 \text{ Nm}^{-1}$ Using $\text{EPE} = \frac{1}{2} kx^2$ Additional $\text{EPE} = \frac{1}{2} k(x_{\text{final}}^2 - x_{\text{initial}}^2) = \frac{1}{2} (100)(0.040^2 - 0.030^2) = 0.035 \text{ J}$
9	C